

T0 Theory: A Unified Physics from a Single Number Summary of the Document Collection

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Abstract

The T0 Theory (Time-Mass Duality) represents a fundamental paradigm shift in theoretical physics. In simple terms: Imagine the universe as a grand puzzle where everything—from the tiniest particles to the vast cosmos—fits together perfectly, without loose ends. The central result of this work is the realization that **all natural constants and physical parameters can be derived from a single dimensionless number**: the universal geometric constant $\xi \approx \frac{4}{3} \times 10^{-4}$. Think of ξ as the “master key” of the universe—a tiny number that emerges from the fundamental shape of three-dimensional space and unlocks explanations for gravity, the speed of light, particle masses, and more.

This collection of over 200 scientific documents systematically develops a complete physical theory that unifies quantum mechanics, relativity, and cosmology—based on the principle of absolute time T_0 and the intrinsic time-field-mass relationship. In everyday language: It’s as if we’re rewriting the rules of physics so that time is stable and reliable (not bendy like in Einstein’s view), while mass can change like sand in the wind, all connected through this elegant geometric idea. The foundational documents follow a purely geometric path, deriving ξ from the three-dimensional structure of space and constructing all other constants from it, including the fine-structure constant $\alpha \approx 1/137$, particle masses, and coupling strengths, without introducing additional free parameters. No more arbitrary numbers; everything flows from a single simple source, making the universe feel less random and more like a beautifully designed whole. Remarkably, the theory postulates a static universe without expansion, as detailed in the CMB document, thereby rendering concepts like dark matter or dark energy superfluous.

.1 The Core Principle: Everything from One Number

The fundamental insight of the T0 Theory can be summarized in one sentence:

Central Theorem of the T0 Theory

All physical constants—gravitational constant G , Planck constant $\hbar$, speed of light c , elementary charge e , as well as all particle masses and coupling constants—can be mathematically derived from a single dimensionless number: the universal geometric constant

$$\xi = \frac{4}{3} \times 10^{-4},$$

which emerges from the fundamental three-dimensional space geometry (harmonic fourth 4:3, QFT loop scaling 10^{-4}). From ξ follows the fine-structure constant as:

$$\alpha = \xi E_0^2 \approx \frac{1}{137.04},$$

where α serves as a secondary electromagnetic coupling without primacy.

In everyday language, this means: We've reduced the "why" of physics to a single, space-born number—no magic, just geometry doing the heavy lifting.

.2 Foundations of the T0 Theory

Time-Mass Duality

In contrast to standard physics, where time is relative and mass is constant, the T0 Theory postulates:

- **Absolute Time Measure** T_0 : Time flows uniformly everywhere in the universe—like a universal clock that ticks the same for everyone, no matter where you are.
- **Variable Mass**: Mass varies with the energy content of the vacuum—think of mass as flexible, changing depending on the "hum" of the empty space around it.
- **Intrinsic Time Field** $T(x, t)$: Every particle carries its own time field—each building block of matter has its personal timer that influences its behavior.

The fundamental relationship is:

$$m(x) = \frac{\hbar}{c^2 T(x, t)(x)} = m_0 \cdot (1 + \kappa \Phi(x)),$$

where κ is traceable to ξ via geometric scaling. Mathematically, this duality treats time and mass as variables, ensuring the framework remains fully compatible with established mathematical structures while enabling a unified description of physical phenomena. Simply put: By letting time and mass dance as adjustable partners, we keep the math clean and intuitive, connecting old ideas with new ones without sacrificing a weld.

The Parameter ξ

The central parameter of the theory is:

$$\xi = \frac{4}{3} \times 10^{-4},$$

a purely geometric construct from 3D space that connects quantum mechanics with gravity. This parameter encodes the fundamental coupling between energy and spatial structure, from which all hierarchies emerge. It's like the ratio that tells space how to "scale" energy—small but mighty, whispering the secrets of why electrons are light and protons heavy.

.3 Derivation of All Natural Constants

Everything Follows from ξ

The T0 Theory demonstrates that:

1. **Gravitational Constant:**

$$G = f_G(\xi, m_P, c, \hbar),$$

where all inputs are reducible to ξ -scaled geometric units. Gravity? Just a wave from the geometry of space, tuned by ξ .

2. **Particle Masses** (Electron, Muon, Tau, Quarks): The particle masses follow a universal scaling law analogous to the ordering principles of atomic energy levels, where quantum numbers (n, l, j) dictate hierarchical structures in a manner similar to atomic shells and subshells—imagine particles stacked like floors in a building, each level set by simple rules, much like electrons orbiting atoms. Thus,

$$\frac{m_e}{m_P} = g(\xi), \quad \frac{m_\mu}{m_e} = h(\xi), \quad \frac{m_\tau}{m_\mu} = k(\xi),$$

via universal scaling laws $\xi_i = \xi \times f(n_i, l_i, j_i)$. No more guessing why some particles are 200 times heavier; it's all patterned like a cosmic family tree.

3. **Coupling Constants** (electroweak, strong, electromagnetic):

$$\alpha_W = f_W(\xi), \quad \alpha_s = f_s(\xi), \quad \alpha = f_\alpha(\xi).$$

These “strengths” of forces? Derived like branches from the same geometric trunk.

4. **Cosmological Parameters:** Static universe metrics and CMB temperature $T_{\text{CMB}} = f_{\text{CMB}}(\xi)$, with redshift mechanisms derived from time-field variations (see CMB document for detailed explanation without expansion).

(Note of 2 Oct. 2026 on notation: here $f(n, l, j)$ is the geometric structure factor of the quantum numbers, not a frequency and not the base winding number $f = 1/\xi$. Cf. Doc. 190, R147.)

.4 Experimental Predictions

The T0 Theory makes precise, testable predictions:

Concrete Predictions

- **Anomalous Magnetic Moment:** $(g - 2)_\mu$ calculation solely from ξ —a quirky electron-like wobble explained without extras.
- **Koide Formula:** Exact mass relation of leptons via ξ -scaling—the math that ties the weights of three particles in a neat loop.
- **Redshift:** Modified interpretation without expansion, governed by ξ —why distant stars look “stretched” without the universe inflating.
- **CMB Anisotropies:** Explanation through time-field variations rooted in ξ —the microwave “echo” of the cosmos as geometric echoes.

These are no wild guesses; they are verifiable with today’s labs and invite everyone—physicists or curious minds—to put the theory to the test.

.5 Structure of the Document Collection

This collection encompasses:

- **Foundations:** Mathematical formulation of time-mass duality under ξ -geometry—the basics, explained step by step.
- **Quantum Mechanics:** Deterministic interpretation, Bell inequalities—quantum weirdness made predictable and local.
- **Quantum Field Theory:** Lagrangian formalism in the T0 framework—fields dancing to a unified tune.
- **Cosmology:** Static universe, redshift, CMB—a stable universe that still surprises, without expansion, dark matter, or dark energy.
- **Particle Physics:** Mass spectrum, anomalous moments, Koide formula—the particle zoo, tamed.
- **Technical Applications:** Photon chip, RSA cryptography—real tricks from the theory.
- **Experimental Tests:** Verifiable predictions—hands-on ways to probe the ideas.

Note: The documents consistently follow the geometric ξ -path, deriving all physics from 3D space principles, with α and other constants appearing as emergent features. We've woven in plain language throughout so non-experts can dive in without drowning in jargon.